



REVIEW OF EQUATION OF A STRAIGHT LINE & ERROR



LEARNING OBJECTIVES

- Discuss what straight lines are and why they matter.
- Simplify complex equations by straightening them out.
- Explain how to create graphs in Physics and what they reveal.
- Apply simple math to draw graphs that represent real-life situations.
- Explain the importance of error analysis in Physics practical.





LINEAR EQUATION (EQUATION OF A STRAIGHT LINE)

Definition:

The equation of a straight line is a mathematical representation that describes a linear relationship between two variables.

Equation of a straight line is given as $y = mx + c$

Where:

y = the dependent variable (usually the vertical axis on the graph)

x = the independent variable (usually the horizontal axis on the graph)

m = the slope (gradient) of the line i.e. the rate of change of y wrt x

c = the y intercept (the value of y when $x = 0$)



LINEAR EQUATION (EQUATION OF A STRAIGHT LINE)

- y intercept: The value of y when $x = 0$
 $y = mx + c$
 $\Rightarrow c = y \text{ intercept}$
- x intercept: The value of x when $y = 0$.
- If $y = 0$ and $y = mx + c$, then $0 = mx + c$
 $x = (-c)/m$
 $x = -(y \text{ intercept})/\text{slope}$

Examples of linear equations are:

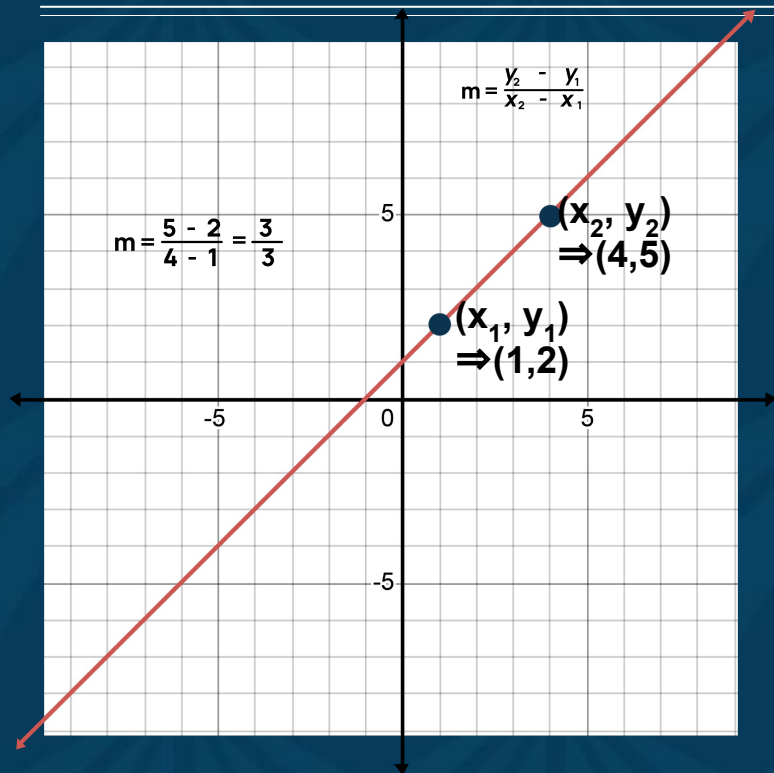
$$y = 24x + 16$$

$$7y - 40x = 35$$





LINEAR EQUATION (GRAPHING)





TECHNIQUES FOR LINEARIZING NONLINEAR EQUATIONS

- Logarithmic transformation:
 - Useful for dealing with exponential growth or decay phenomena.
 - Taking the logarithm of a nonlinear equation can convert it into a linear equation.

Example:

Linearizing exponential decay equation: $N = N_0 e^{-\lambda t}$ to a linear form using logarithms.



TECHNIQUES FOR LINEARIZING NONLINEAR EQUATIONS

Exponential Decay:

$$N = N_0 e^{-\lambda t}$$

To linearise, we take the natural log of both sides of the equation, we have:

$$\ln(N) = \ln(N_0) - \lambda t$$

Rearrange

$$\ln(N) = -\lambda t + \ln(N_0)$$

Compare with $y = mx + c$, if $\ln(N)$ is to be plotted against t , then

slope, $m = -\lambda$, y - intercept = $\ln(N_0)$



TECHNIQUES FOR LINEARIZING NONLINEAR EQUATIONS

- Power-law transformation:
 - Suitable for equations with power-law relationships, where one variable is raised to a power.
 - Taking logarithms of both sides can convert the equation into a linear form.

Example: Linearizing power-law equation - converting $y = kx^n$ to a linear form using logarithms.



TECHNIQUES FOR LINEARIZING NONLINEAR EQUATIONS

Power Law:

$$y = kx^n$$

To linearise, we take the log of both sides of the equation,
we have:

$$\begin{aligned} \log(y) &= \log(k) + n \log(x) \\ \text{Rearrange } \rightarrow \log(y) &= n \log(x) + \log(k) \end{aligned}$$

Compare with $y = mx + c$ (plot $\log(Y)$ against $\log(x)$)
slope, $m = n$, y -intercept = $\log(k)$



MID-LESSON QUESTION

Given that resistance of an ohmic conductor is $R = \frac{\rho L}{A}$
where L and A are the length, and the cross-sectional area of the conductor respectively. ρ is the resistivity of the conductor, which depends on the type of the conductor's material and the temperature.

What will the slope be if we plot :

Graph of R vs L ,

Graph of R vs $\frac{1}{A}$,

Graph of R vs ρL ,

Graph of RA vs L



ANSWER TO MID-LESSON QUESTION

Given that resistance of an ohmic conductor is $R = \frac{\rho L}{A}$

where L and A are the length, and the cross-sectional area of the conductor respectively. ρ is the resistivity of the conductor, which depends on the type of the conductor's material and the temperature.

What will the slope be if we plot :

Graph of R vs L , then $\frac{\rho}{A}$ will be the slope

Graph of R vs $\frac{1}{A}$, then ρL will be the slope

Graph of R vs ρL , then $\frac{1}{A}$ will be the slope

Graph of RA vs L , then ρ will be the slope



TECHNIQUES FOR LINEARIZING NONLINEAR EQUATIONS

		$Y = mx + c$			
Equation	Linearized Form	y-axis	x-axis	Slope, m	Intercept, c
Simple Pendulum: $T = 2\pi\sqrt{L/g}$	$T^2 = 4\pi^2(L/g)$	T^2	L	$4\pi^2/g$	0
Hooke's Law: $F = kx$	$F = kx$	F	x	k	0
Coulomb's Law: $F = k(q_1q_2/r^2)$	$F = kq_1q_2/r^2$	F	$1/r^2$	kq_1q_2	0



TECHNIQUES FOR LINEARIZING NONLINEAR EQUATIONS

Equation	Linearized Form	$Y = mx + c$			
		y-axis	x-axis	Slope, m	Intercept, c
Boyle's Law $\log(P) = \log(k) - \log(V)$	$\log(P) = \log(k) - \log(V)$	$\log(P)$	$\log(V)$	-1	$\log(k)$
Exponential Decay: $N = N_0 e^{-\lambda t}$	$\ln(N) = -\lambda t + \ln(N_0)$	$\ln(N)$ (natural logarithm of N)	t (time)	$-\lambda$	$\ln(N_0)$



BASICS OF GRAPHING IN PHYSICS

Graphing in physics helps in the following ways:

- visualization of data and relationships.
- analysis and interpretation of experimental results.





BASICS OF GRAPHING IN PHYSICS

Guidelines for constructing and interpreting graphs:

- Proper scaling of axes.
- Labeling of axes.
- Plotting data points accurately.
- Drawing the best-fit line or curve.



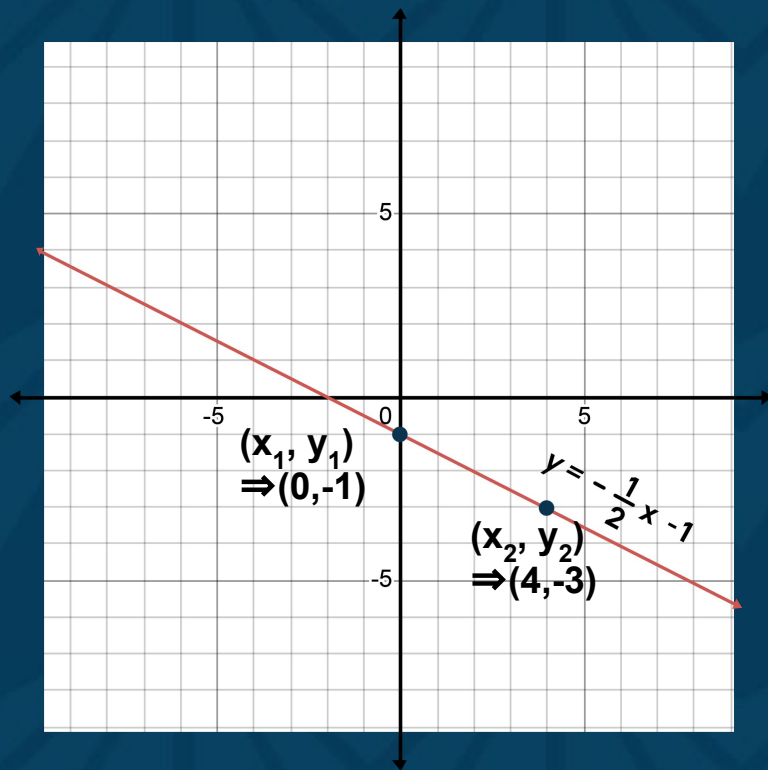
BASICS OF GRAPHING IN PHYSICS

- Relationship between linear equations and straight line graphs:
 - Linear equations generate straight line graphs.
 - Slope represents the rate of change or gradient of the graph.





BASICS OF GRAPHING IN PHYSICS





WHAT IS ERROR IN PHYSICS PRACTICAL?

- Error is defined as the difference between a measured value and the true value.
- Sources of errors in Practical Physics include:
 - Instrument errors
 - Observation error
 - human error, etc.





WHAT IS ERROR IN PHYSICS PRACTICAL?

Errors can be either systematic or random.

- Systematic errors are caused by a consistent failure of an instrument
- Random errors are caused by factors that are not predictable or consistent.



PERCENTAGE ERROR

- The actual error is the amount by which an experimental value differs from the true value.
- For example, if a man measures the length of a rope to be 6.0m instead of 6.3m. The actual error in the measurement is 0.3m i.e. 6.3-6.0.



The relative or fractional error = $\frac{\text{Actual Error}}{\text{True Value}}$

Percentage Error = $\frac{\text{Actual Error}}{\text{True Value}} \times 100\%$



PERCENTAGE DIFFERENCE

Percentage difference = $(\text{observed value} - \text{Reference value}) / \text{Reference value}$

A positive percentage difference indicates that the observed value is higher than the reference value, while a negative percentage difference indicates that the observed value is lower.

The percentage difference is commonly used in various fields, including physics, chemistry, economics, and data analysis.



CONCLUSION

- Understanding the equation of a straight line, learning how to linearize nonlinear equations, and mastering the basics of graphing in physics are essential skills for analysing and interpreting physical phenomena.
- These skills allow you to effectively model relationships, make predictions, and draw meaningful conclusions from experimental data.



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THANK YOU